CIVL 1100
Discovering Civil and Environment Engineering

Geotechnical Lab Session

- Laboratory Soil Liquefaction Tests

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Outline

- Objectives and test arrangement
- Introduction to soil liquefaction
- Test programme
- Test procedures
- Lab report

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Objectives and test arrangement

- Objectives
  - To gain insights into soil liquefaction
  - To identify key factors that influence soil liquefaction
- Test arrangement
  - Preparation of sand-bed model
  - Earthquake shaking to trigger soil liquefaction

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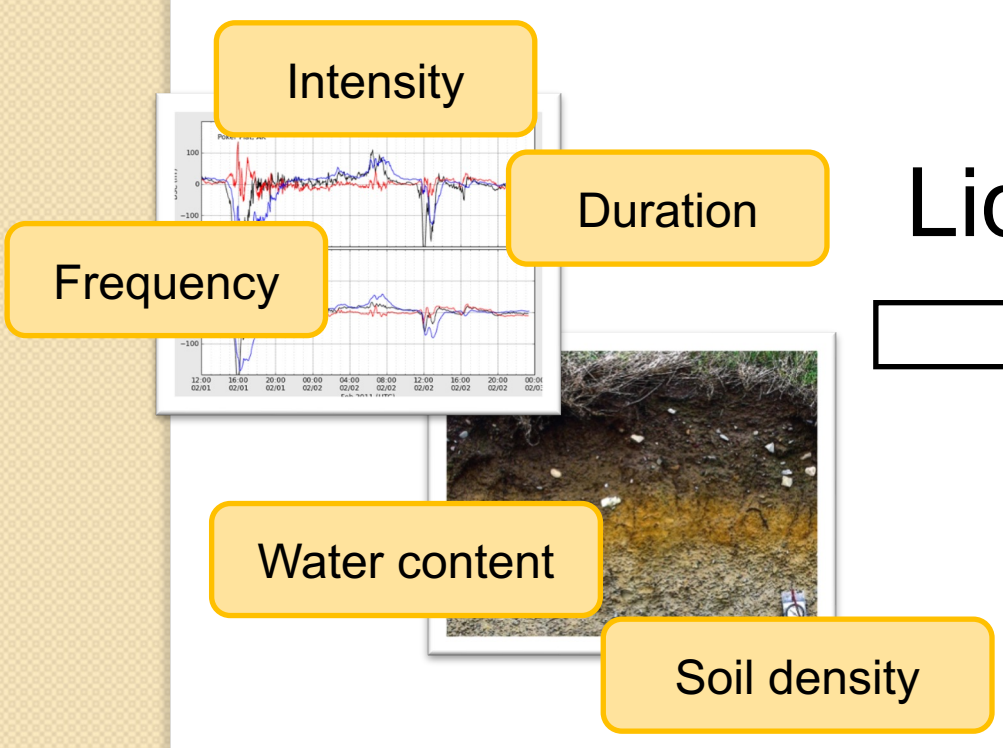
Introduction to soil liquefaction

- What is soil liquefaction?



Liquefy

Liquefaction – The phenomenon that saturated soil behaves like a liquid when shaken by an earthquake

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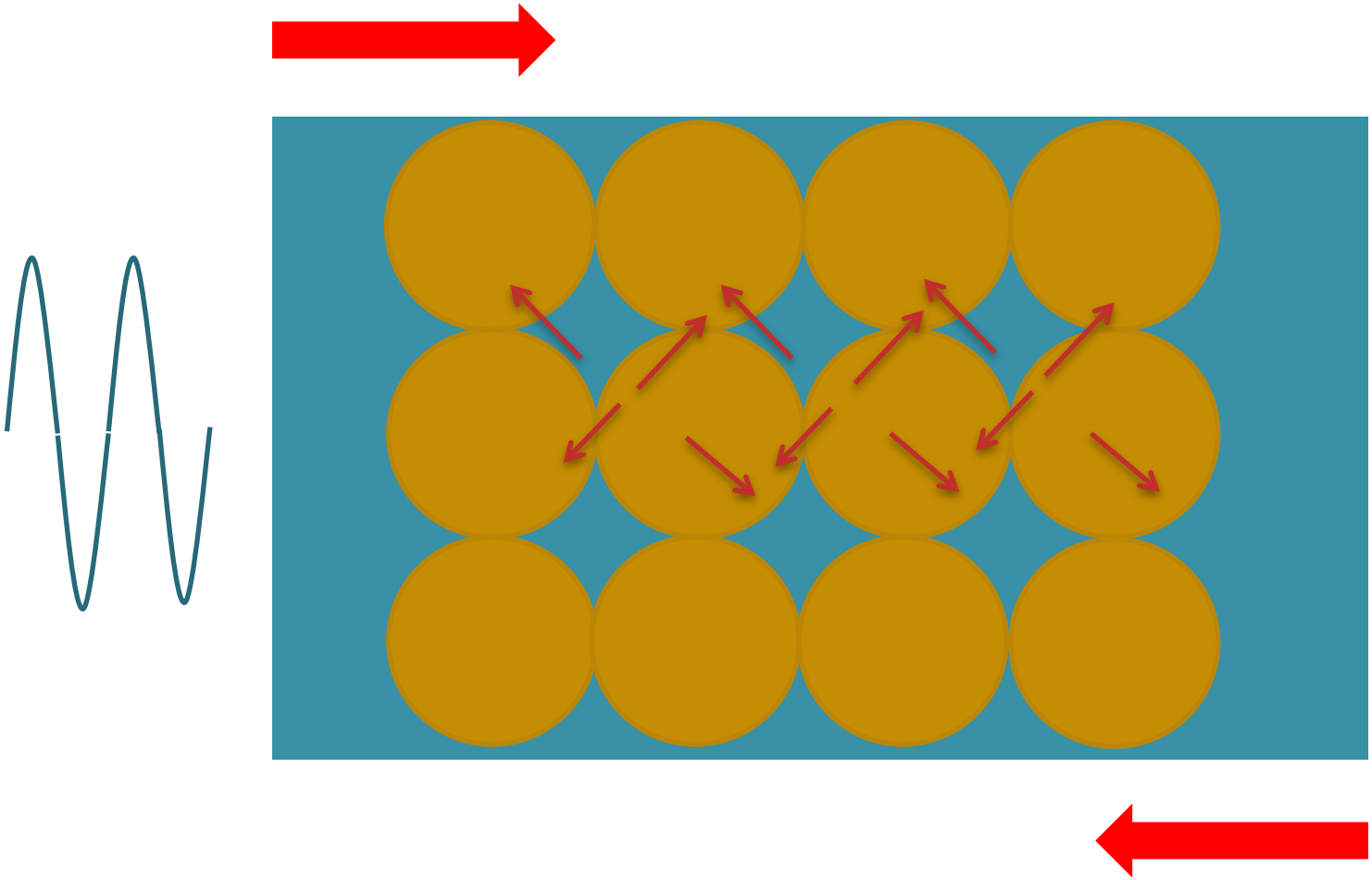
Introduction to soil liquefaction

- Mechanism of earthquake induced soil liquefaction



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Introduction to soil liquefaction

- Typical effects of soil liquefaction

- Loss of bearing capacity

The ground loses all shear strength
→ Not able to support superstructures

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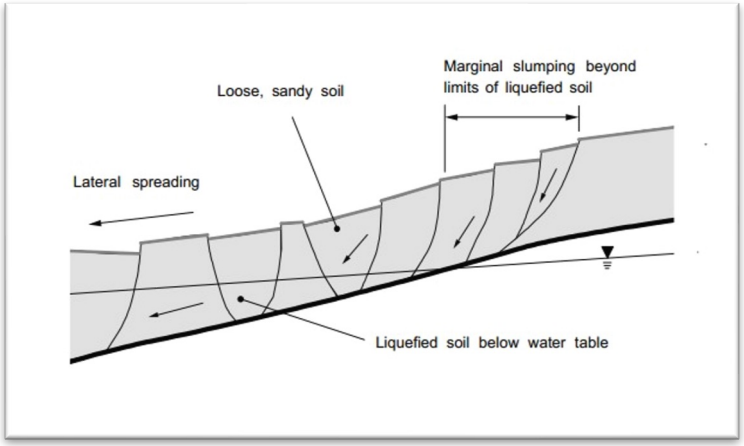
Introduction to soil liquefaction

- Typical effects of soil liquefaction

- Lateral spreading



Lateral displacement due to the ground sliding down very gentle slopes or sliding towards stream banks riding on a buried liquefied layer

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Introduction to soil liquefaction

- Typical effects of soil liquefaction

- Sand boils

Sand-laden water can be ejected from a buried liquefied layer and erupt to the surface to form “sand volcanoes”; the surrounding ground often fractures and settles

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Test programme

Test	Water (mL)	Intensity (m/s^2)	Slope
1	2400	9	N
2	2600	9	N
3	2600	9	Y
4	2600	12	Y

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Test procedures

- **Sample preparation**
- Earthquake shaking

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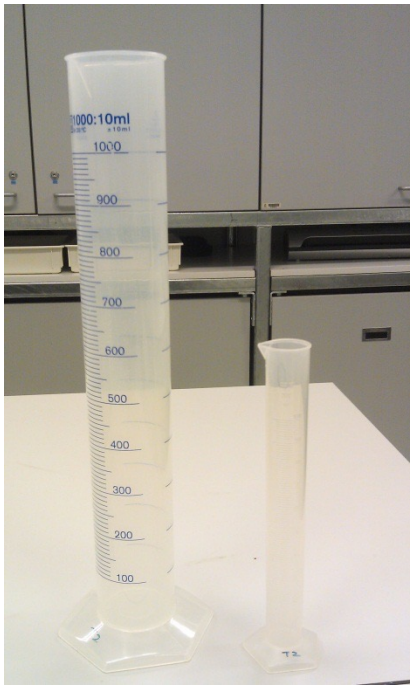
Test procedures: Sample preparation

1. Toyoura Sand

2. Water



3. Scoops

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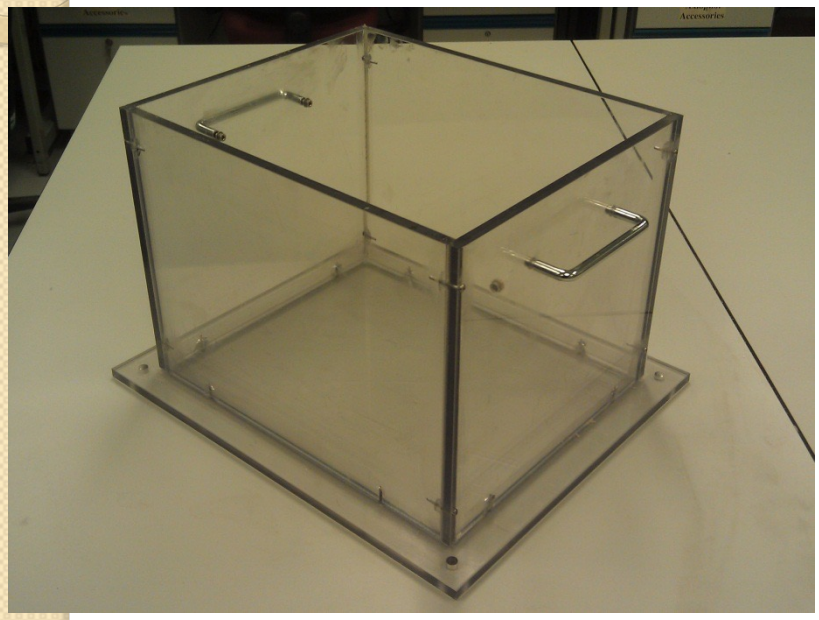
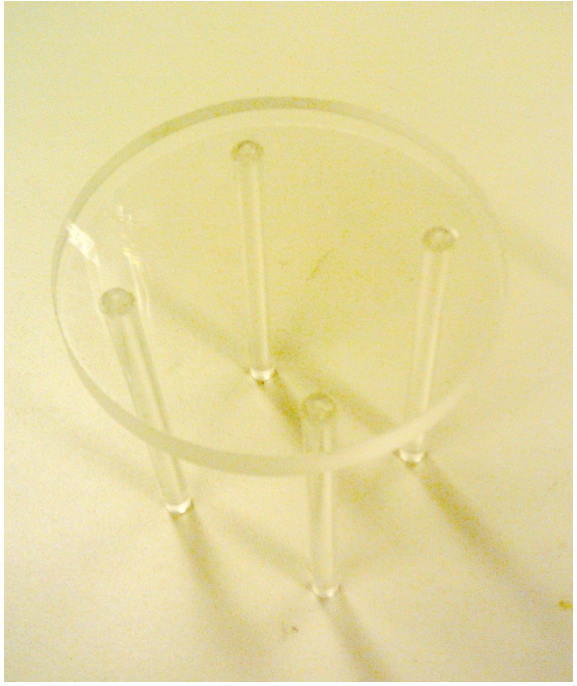
Test procedures: Sample preparation



4. Perspex soil container



5. Pile foundation model

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Test procedures: Sample preparation

1. Weight 10 kg of Toyoura sand
2. Take proper amount of water using the measuring cylinder
3. Pour sand into the Perspex soil container
4. Add water and mix sand with water uniformly. Keep the sample in a loose state (do not compact!)

   (Around 8-10 scoops of sand with 500 mL of water)
5. Measure the dimensions of the sample and calculate its volume
6. Insert the pile foundation model into the sand bed or the slope

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Test procedures

- Sample preparation
- **Earthquake shaking**

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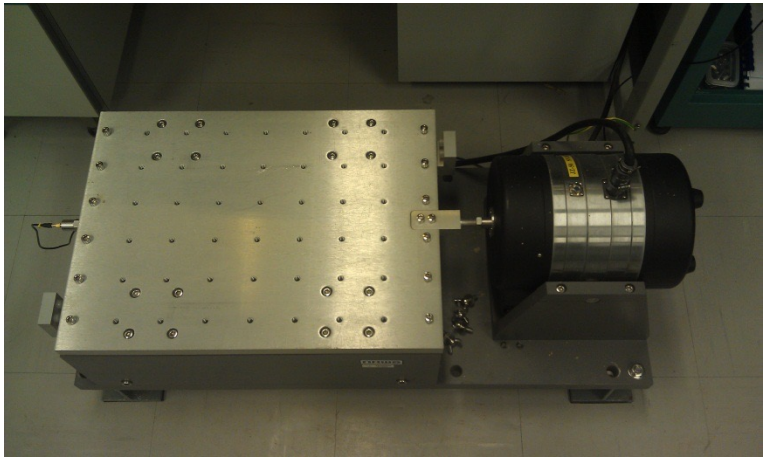
Test procedures: Earthquake shaking

NAWS-Z30 earthquake shaking table

- To simulate earthquake loading



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Test procedures: Earthquake shaking

1. Fix the Perspex soil container on the shaking table
2. Tune the shaking amplitude and frequency
3. Observe and record soil and structure behaviours on your own (record the shaking process if necessary)
4. Loosen the Perspex soil container and clean it after the shaking test

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Lab report

- The template for the lab report is available on CANVAS
- Each group must prepare one lab report and everyone in the group must submit the prepared lab report
- You should finish all questions in the lab report
- The lab report is due:
- **11:59 pm 07/05/2024 (Tue) – For sessions on April 30**
- **11:59 pm 14/07/2024 (Tue) – For sessions on May 7**
- All late submittions and plagiarized works will be subjected to penalties

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